

Boundary buffering: limiting interference and information leakage in design teams

Boundary
buffering in
automotive
industry

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Abstract

Purpose – Two challenges faced by automotive component design projects within contracted design agencies are (1) specification changes requested by the manufacturers and (2) product information or core technology knowledge leakage to external actors. We examine the effects of targeted boundary activities that address these challenges under the contingencies of environmental uncertainty and project complexity.

Design/methodology/approach – Based on Boundary management theory, a bidirectional model of boundary buffering was conceptualized in the context of design agency teams developing automotive components. A survey is derived from the proposed model. Regression analysis is performed using empirical data from 234 auto component design projects in Chinese design agencies.

Findings – Boundary buffering activities that strengthen outside-in boundaries and inside-out boundaries directly improve the final design quality. Further, the magnitude of effect for outside-in buffering on design quality is enhanced under environmental uncertainty, while the impact of inside-out buffering on design quality is enhanced under project complexity.

Research limitations/implications – Boundary activities should consider differences in boundary targets, directional flow of information, and context of scope.

Practical implications – Automotive component design agents should attend to both outside-in and inside-out boundary buffering, especially under conditions of environmental uncertainty or project complexity.

Originality/value – The proposed bidirectional view on boundary buffering adds perspective to team boundary management theory. Specific contingencies include common risk elements of project complexity and environmental uncertainty not typically associated with the need for buffering activities.

Keywords Automotive industry, New product design, Design agents, Boundary buffering, Environmental uncertainty, Project complexity

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1. Introduction

Automotive manufacturers increasingly outsource component design to specialized design agents, an arrangement that alters relationships in the traditional buyer–supplier dyad (Lechler *et al.*, 2019; Wilhelm and Dolfmsma, 2018; Gong *et al.*, 2018; Wilhelm *et al.*, 2016; Ateş *et al.*, 2015; Gray and Handley, 2015). For example, increasing foreign automotive component companies set up Sino–foreign joint ventures on automotive component design in China to serve local manufacturers, which occupy more than 70 percent of the auto-part market share (Export.gov, 2019). Design agents differ from traditional suppliers that provide standard or low-tech goods because design agents assume the responsibility to fulfill the component design task for the buying firm and integrate any second-tier suppliers (Lechler *et al.*, 2019; Wilhelm and Dolfmsma, 2018; Wilhelm *et al.*, 2016). Organizationally, the design agency initiates the design project and assembles a team with the essential talent to fulfill contractual obligations. The design project team faces two significant difficulties unique to the resulting triad: (Ancona and Caldwell, 1988) component specification changes requested by the manufacturers throughout the project, and (Ancona and Caldwell, 1990) leakage of product information and core technology knowledge to inappropriate external actors. The challenge is to overcome these difficulties.

The first difficulty of design changes is due to the buyer holding approval authority, even though the locus of design control transfers to the design agent. The buyer expects flexibility to change the requirements in response to market changes and other uncertainties (Ateş *et al.*, 2015; Stevenson and Spring, 2009). This flexibility entails extensive investment and operating overhead for the design agency, which may drain resources and alter talent requirements of the affected design project team and the design agency in general (Kalaitzi *et al.*, 2019; Gibson and Dibble, 2013). Thus, the design project team must defend their inbound boundary to protect the project from the interference of the buyer in terms of managing design change requests (Chakkol *et al.*, 2018). The second challenge is that the design project is, by nature, an innovation activity requiring the application of knowledge from internal and external sources (Gong *et al.*, 2018; Wilhelm and Dolfmsma, 2018). In this environment, the team must keep product information and core technology knowledge from leaking to inappropriate external actors (Lawson *et al.*, 2012). Frequent contact with the buyer, other design teams, and suppliers adds complexity to the project and increases the difficulties of controlling leakage. Thus, the project team must defend against the outbound boundary to prevent confidential information and knowledge leaks. In summary, Figure 1 highlights the potential leaks and interference in this unique context.

From the theoretical perspective, a boundary in a social system is more than a wall separating the system from its environment; it is the domain of effort through which the system interacts with its environment to survive (Yan and Louis, 1999). Boundary management theory, therefore, considers an organizational boundary as both a point of exchange (boundary spanning) and a demarcation (boundary strengthening) (Kalaitzi *et al.*, 2019; Wilhelm and Dolfmsma, 2018; Faraj and Yan, 2009). The former stresses the perspective of resource dependence to interact for exchange across the boundaries (Gong *et al.*, 2018; Bode *et al.*, 2011; Yan and Louis, 1999). Studies enumerate the positive impact of boundary spanning on team performance (Wilhelm and Dolfmsma, 2018; Marrone, 2010). On the other hand, corresponding to the view of information processing (Bode *et al.*, 2011), boundary demarcation highlights two focuses: (Ancona and Caldwell, 1988) boundary reinforcement, focusing on defining borders around the team to develop an identity and nurture a supportive climate (Yan and Louis, 1999); and (Ancona and Caldwell, 1990) boundary buffering, protecting the project team from external interference and resource leakage (Kalaitzi *et al.*, 2019). This study focuses on the boundary buffering activities in the automobile design agency context.

Boundary buffering may be defined as the “activity to insulate the system from unwanted exposure to the environment, including preventing disturbances and uncertainties, negative

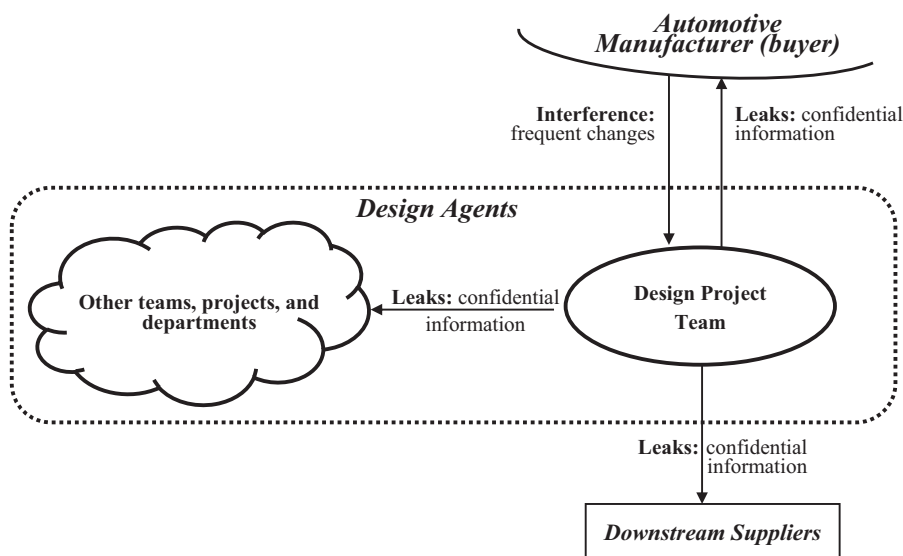


Figure 1.
Leak and interference potential for a design project team

input caused by undesired access to the team boundaries, and leakage of valuable resources” (Somech and Khalaili, 2014, p. 278). Empirical studies consider activities controlling and filtering the information flow and generally consider the activities as a single-dimensional concept (Kalaitzi *et al.*, 2019; Dey and Ganesh, 2017), although boundary buffering activities include both “inbound” (e.g. filtering the inputs) and “outbound” (filtering the outputs) requirements. This limitation could be misleading because (Ancona and Caldwell, 1988) an incorrect connotation makes it difficult to compare results across studies or integrate the existing literature, and (Ancona and Caldwell, 1990) blurs the development of practical activity and construct development, limiting the usefulness of buffering for scholars and practitioners.

Interestingly, recent studies in other disciplines propose a bidirectional view to examine distinct “inbound” and “outbound” flows (Subramanian *et al.*, 2019; Cassiman and Valentini, 2016). For example, the “inflows” of knowledge and “outflows” of knowledge are found to both accelerate internal innovation in R&D productivity (Cassiman and Valentini, 2016). We argue this bidirectional view guides investigation of the effect of boundary buffering in the context of the automobile component design projects. Specifically, we classify boundary buffering into (Ancona and Caldwell, 1988) the “outside-in buffering (OIB)” defined as the effort of the design project team to protect the team members from the outside-in interference, and (Ancona and Caldwell, 1990) “inside-out buffering (IOB)” defined as the effort of the design project team to prevent inside confidential information from leaking to outside actors (Ancona and Caldwell, 1990). These two types of buffering activity should be separately examined because: 1) they have different boundary targets: OIB targets the external buyers, but IOB targets the internal project members; and 2) they have different boundary objects: OIB on the changes of the design task and the IOB on the confidential information. Given the importance of boundary buffering on project performance, however, little is known on its effect on automobile design projects of design agents. Therefore, in this study, we first consider: “Whether outside-in boundary buffering and inside-out boundary buffering activities have differing but beneficial, effects on automobile design team performance?”

The effect of boundary buffering is suggested to be context-dependent because contexts would lead the project teams to decide when and how to conduct buffering work (Dey and

Ganesh, 2017). Prior research investigated moderating contextual factors on boundary management, including institutional pressures, requisite variety, and continuous environmental changes (Lynn, 2005). Unfortunately, no study separates the contingencies by activity type or in the design agency triad. External interference in our context is often a result of changes in the environment requiring design change. Thus, we argue that external environment uncertainty is an important contingency, in particular given the heightened volatility in the Chinese automotive component industry (Eloot *et al.*, 2013). Higher environment uncertainty results in frequent changes to component design required by the buyers (Eloot *et al.*, 2013). Given that OIB aims to protect the design project from excessive interference, environment uncertainty should be expected to determine the magnitude of the effect of OIB. Furthermore, project complexity is an essential contingency on the effect of the IOB because (Ancona and Caldwell, 1988) project internal complexity is often observed in the Chinese component industry (Spring, 2016), (Ancona and Caldwell, 1990) information leakage risk is caused by internal project complexity (Cooper *et al.*, 1998; Geraldi *et al.*, 2011; Maylor and Turner, 2017), and (Ateş *et al.*, 2015) IOB is specially developed to prevent information leakage. Thus, we raise a second research question: “*Would the environmental uncertainty and project complexity impact the effects of the OIB activities and IOB activities on design team performance?*”

To answer the research questions, this study adopts boundary management theory to propose and test a research model to examine whether there is a positive relationship between the different buffering activities and project team performance and whether environment uncertainty and project internal complexity moderate the relationship. In our case, we argue project team performance to be narrowly defined as the component design quality. Results from 234 auto component design projects in 124 automobile design agencies reveal that both OIB and IOB are positively associated with final design quality, that OIB activity contributes to the final design quality more significantly when environmental uncertainty is high, and that IOB activity best increases design quality when project complexity is high. The findings contribute to the literature in several ways: (Ancona and Caldwell, 1988) by providing empirical evidence to better understand the phenomenon of the under-explored supply chain context of having a design agency as a third party instead of the traditional buyer–supplier dyad (Ateş *et al.*, 2015), (Ancona and Caldwell, 1990) by providing new insight to our understanding of boundary management activities by adopting a bidirectional view, and (Ateş *et al.*, 2015) by providing a new insight to our understanding of boundary management theory by identifying specific context contingencies (Faraj and Yan, 2009).

2. Foundation and hypotheses

2.1 Boundary management

Boundary management theory proposes boundary spanning as well as boundary strengthening activities to establish, maintain boundaries, and manage interactions across those boundaries to improve an organization’s performance (Dey and Ganesh, 2017). These two kinds of boundary management activities respectively capture the positive and negative impact of external communication, which correspond to two views of the boundary as a point of exchange or demarcation (Yan and Louis, 1999). In new product development (NPD) work in the project literature, agile project management identifies with a positive impact of boundary spanning (Chakkol *et al.*, 2018; Dey and Ganesh, 2017; Prajogo and Sohal, 2013). The concurrent engineering of manufacturing in the automotive sector benefits from external communication within automotive supply chains and across functional teams to minimize time-to-market (Correia *et al.*, 2014). Interestingly, researchers observe that the boundary strengthening activities that help the project define and defend its borders are less studied (Dey and Ganesh, 2017), especially when excessive communication might harm the project team’s success (Gibson and Dibble, 2013).

Around the role of the boundary as a demarcation, the literature describes strengthening and tightening activities, including protecting, guarding, and reinforcement. Boundary tightening and strengthening are two synonymous concepts in the boundary management literature (Somech and Khalaili, 2014; Dey and Ganesh, 2017). Dey and Ganesh (2017) divide strengthening into buffering and reinforcement. Buffering enables teams to block external disturbances and prevent leakage of information and other resources necessary to execute the innovative product (Yan and Louis, 1999). The reinforcement of borders creates a unique sense of belonging and team identity (Yan and Louis, 1999). Ancona and Caldwell (1990; 1988) suggest guarding and protecting as two sub-components under buffering, where guarding is inward-focused and protecting is outward-focused. Protecting as one kind of tool of ambassadorial activities could limit external pressures and interference (Ancona and Caldwell, 1990). On the other hand, guarding involves “monitoring the information and resources that others request from the group and determining how the group will respond to those demands” (Ancona and Caldwell, 1988, p. 478). A guarding activity is to judge the legitimacy of an external request. When the request is deemed legitimate, then the required information or resources will be delivered. Guarding is also conceptualized as “keeping information within the team’s boundaries in order to protect the team or present a specific image of the team to outsiders” (Ancona and Caldwell, 1990, p. 127).

In order to examine buffering activity in more detail, we suggest bidirectional boundary buffering as the basis for addressing issues of information leakage and external interference for our context. We summarize both directions, OIB and IOB, in our context across the situational expectations of boundary management theories in Table I. According to theory, at least six fundamental properties should be present in the context when studying boundary activity. First, the environmental constraints of external disturbances and internal demands create the motivation for conducting boundary activity (Fennell and Alexander, 1987). In our case, environmental uncertainty motivates OIB, and project complexity motivates IOB. Second, sources of constraints indicate targets of boundary activity (Dey and Ganesh, 2017; Marrone, 2010) and the boundary objects which interact with the targets (Koskinen and Mäkinen, 2009; Iorio and Taylor, 2014). In this study, environmental uncertainty results in the buyer adjusting component design and project complexity will create more opportunities for team members to leak confidential information. Therefore, buyers will be the targets of OIB, and project team members will be the target of IOB. Third, changed information of documents, models, specifications, and prototypes of the auto component design will be the objects of OIB, and core technical knowledge of the design project will be the objects of IOB. Fourth, the boundary objects determine the connotation of the boundary, including the physical, temporal, and cognitive (Kreiner *et al.*, 2006). Both OIB and IOB are information connotations considering their objects. Fifth, specific boundary functions focus on specific targets and objects (Marrone, 2010). In this study, the OIB protects the project members from outside-in interference, and IOB prevents confidential information leakage. Sixth, responsibility for boundary management is delegated to a lead actor (Marrone, 2010). In this study, the project manager assumes responsibility for both OIB and IOB.

Project management scholars emphasize the temporary nature of projects in boundary activities (Lundin and Söderholm, 1995). For example, a temporary nature suggests the project must deliver the expected product or service within a specific time and budget. The limitation of time and budget requires boundary strengthening to focus the project on the planned task (Lundin and Söderholm, 1995), which provides support for the necessity of OIB. Also, the temporary nature is critical in the selection and construction of design quality as a dependent variable. Contracts between buyer and design agency allow flexibility in order to respond to product changes, with the buyer holding final acceptance rights (Elloot *et al.*, 2013). When taking the temporary nature of projects into account, the design project must meet the scope flexibility under a defined schedule. The buyer will determine at the close of the project

Table I.
Six properties of
boundary activity

Situational properties	General definition	Relevance to context
Environmental constraints	External disturbances and dependencies require active responses (Fennell and Alexander, 1987) or internal demands	OIB: External interference due to environment uncertainty, e.g. local manufacturer requiring the adjustments of component design project scope IOB: Confidential information leaking due to project complexity, e.g. a variety of talent and tasks in joint component design
Boundary targets	The actors external to the system requiring boundary management activity (Marrone, 2010)	Targets of IOB: Internal information leakers among project members Targets of OIB: External disturbances of the buyer firm
Boundary objects	Physical objects or abstract concepts around which knowledge sharing, collaboration, cooperation, and communication between the two actors occur (Koskinen and Mäkinen, 2009; Iorio and Taylor, 2014)	Objects of OIB: include changed information in documents, models, specifications, and prototypes of the auto-component design Objects of IOB: Confidential product information and core technology knowledge
Boundary connotation	Physical, temporal, cognitive, limits that define domains as separate from one another and define components within domains (Kreiner et al., 2006, p. 1318)	OIB: Inbound information flow demarcation, located between the design agency and the buyer (inter-organizational boundary) IOB: Outbound information flow demarcation, which may also be located both in the design agency or outside the design agency
Boundary functions	Types of boundary activities conducted to attain the objectives (Marrone, 2010)	OIB: Protect the design project from outside interference activity IOB: Prevent confidential information leakage
Boundary manager	The lead actor is responsible for boundary management (Marrone, 2010)	The component design project manager takes the roles of buffering both the outside-in and inside-out information flow

whether to accept the design as meeting requirements of scope and time. Therefore, the design quality of the component is operationalized as the extent to which the final design meets the requirements of the buyer within the schedule limitation. Responding to changes could enhance the design quality to meet the buyers' needs, but meanwhile may require additional time. As such, the design agency will underinvest in quality unless the buyer provides adequate incentives to enhance their quality-related effort (Handley and Gray, 2015). Therefore, from the design agency perspective, responding to changes may imply a quality decrement at the risk of schedule slippage and non-acceptance. Frequent interference from the buyer or other forces limits the chances of successful delivery.

2.2 External uncertainty and internal complexity

We situate our theorizing within the context of automobile component development projects. The unit of analysis is the new product design team considered at the project level. The design team is responsible for design success and manages the boundaries with external targets, including the buyer, potential suppliers, and other groups within the design agency. The project team must protect against outside interference due to change requests of the buyer and leakage of confidential information and limited resources to actors outside the

team. Uncertain environments and markets logically lead to changes in the specifications of the product sparked by the evolving market perceptions of the buyer or parts availability of downstream suppliers. Project complexity leads to a higher potential for leakage due to the inability to control the larger, more diverse number of tasks and team members (Liu, 2015). Thus, both external uncertainty and project complexity must be managed to govern the team boundary. We expect that OIB efforts will be more prevailing when the external environment is uncertain, and IOB work will be more fundamental when the design project is more complex.

Both external uncertainty and internal complexity represent challenges in projects. We draw on specific features of Geraldi *et al.*'s (2011) five sources of complexity (structural complexity, uncertainty complexity, dynamic complexity, pace complexity, and sociopolitical complexity) and the element complexity and relationship complexities of Ramasesh and Browning (2014). Specifically, for OIB, boundary targets located in the environment outside the project, we adopt two key sources (uncertainty complexity and dynamic complexity) to represent environmental uncertainty. These two sources capture the uncertainty of the external environment (uncertainty complexity) and the unpredictable changes of the buyer (dynamic uncertainty) as conditions requiring acts of OIB. Concerning IOB, the boundary targets are internal project members, who are varied in talent and task and hold greater interdependencies. As the variety of talent and task increase, along with the relationships due to interdependence, the element and relationship complexity increase (Ramasesh and Browning, 2014).

2.3 Hypotheses development

One crucial boundary function to improve performance is to level or smooth variability in inputs and to impose rationing to protect the delineated territory from environmental penetration (Yan and Louis, 1999). In our context, OIB helps keep the design team insulated from frequent shifts in specifications of the buyer. Avoiding outside interference allows the design team to focus on project tasks and reach the desired component quality. For example, during the design project, the buyer might require a change in the original specifications (such as adding a new attribute for the auto component) based on market and customer preference changes (Stevenson and Spring, 2009). If the design team negotiates with the buyer to revise the agreement, tasks are rescheduled and resources reallocated. Plan changes must be made under political and infrastructure constraints and adjusted to accommodate economic and technology windows of opportunity. Such activities may be regrettable. Although the changes may satisfy constituent needs, they may compromise the overall intent or vision of the product or service and drain limited resources (Gibson and Dibble, 2013). Thus, OIB in the context of this study serves to strengthen the auto component project team boundaries against external interference and creates a protected internal atmosphere for the members, thereby contributing to the achieved design quality. Thus, we propose:

H1. OIB has a positive impact on the final component design quality.

External environmental uncertainty captures the external sources of project complexity, including uncertainty and dynamics (Geraldi *et al.*, 2011). Changes in the external environment (e.g. economic/social/political/technological) might require project plan modifications (Simangunsong *et al.*, 2012). All actors must respond to this interference. The buyer will react to changes in the environment by attempting to press more alterations to the component under development (Chang *et al.*, 2002; Stevenson and Spring, 2009). The flexibility of the buyer is maintained by adding a burden to the design agency (Gibson and Dibble, 2013). Further, the design agency will respond to changes as well, resulting in changes to many ongoing projects, resulting in potential shifts to shared resources, alteration of tasks, and modified schedules. The buffering of a design team will be more critical as the external pressures to change increase.

Responding to the uncertainty and changes allows the design team to maximize value creation but typically means an increment of budget and schedule changes. Given the specifics in any contract, more changes to uncertainty mean deterioration of quality, budget, schedule, or some combination (Handley and Gray, 2015). Thus, OIB should limit the impact of interference due to uncertainty in the external environment. For example, autonomous driving design agents for a manufacturer in China report losses due to satisfying the dynamic requirements of the manufacturers (Research and Markets, 2005). Therefore, environmental uncertainty is an essential contingency for design teams in the conduct of OIB, to deliver a quality design in compliance with the contract. Formally:

H2. External environmental uncertainty positively moderates the relationship between OIB and component design quality.

Another function of the boundary to improve team performance is to seal off the productive core and secure sensitive information, which promotes a competitive advantage or uniquely improves the output (Yan and Louis, 1999). The automotive component development is an innovation project, so it is essential to keep product information from leaking to competitors via direct and indirect actors (Lawson *et al.*, 2012). Delegating too much autonomy for external exchanges might prompt opportunistic and unintentional behaviors that leak technical knowledge or product information, limiting the value of the design (Harmancioglu, 2009). IOB manages autonomy and freedom of outward information exchange to a reasonable level (Yan and Louis, 1999). Specifically, in our context, the design team must manage exchanges external to the project. Too many contacts mean greater knowledge loss and lessened accountability (Speier *et al.*, 2011). The design team must conduct an IOB to monitor to prevent the leakage of confidential information from team members to the buyer, suppliers, potential competitors, and other groups in the design agency to maintain the value of the component design quality. Thus, we propose:

H3. IOB has a positive impact on component design quality.

Project internal complexity involves the variety of elements and relationships required to complete the tasks, schedules, and scope of cross-organizational communication (Ramasesh and Browning, 2014; Liu, 2015). Projects with more expertise variety enable potential leakers in the project to conduct leakage behaviors, especially in cross-organizational activities (Tan *et al.*, 2016). Schedule, talent, and task differences require greater interaction and scope of cross-organizational communication with more external stakeholders (e.g. buyers, suppliers, other competitors), which both expose the project to higher risks of information and knowledge leaking (Maylor and Turner, 2017; Liu *et al.*, 2019). The project suffers a high risk of leakage if there are no effective preventive measures in place (Speier *et al.*, 2011). Frishammar *et al.* (2015) emphasize that protective efforts such as isolating mechanisms must be present in projects of high complexity. Therefore, IOB becomes essential under conditions of higher internal complexity. Formally:

H4. Project internal complexity positively moderates the relationship between IOB and component design quality.

We include control variables for each possible level of analysis for the model under the limitation that they may provide alternative explanations for our hypothesized relations. Alternative explanations mean the control variable may have a potential impact on the dependent variable but also on an independent variable, or may moderate the hypothesized relationships (Spector and Brannick, 2011). Following these criteria, six control variables, shown in Figure 2, are in our research model. Project managers' work age represents the individual level and captures the experience of the project manager as the lead boundary manager and may influence the ability to complete the deliverable as well as choose the better

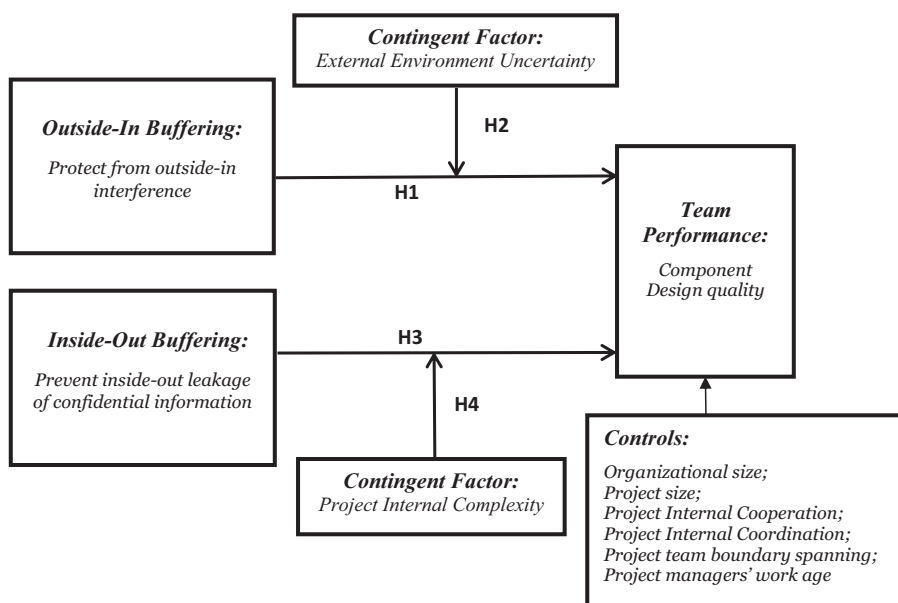


Figure 2.
Research model

boundary actions (Dey and Ganesh, 2017). At the team level, project team boundary spanning has a potential interaction with buffering work on the project performance (Dey and Ganesh, 2017). For the project level, project size influences the actions taken for boundary management and impacts project performance. Project internal coordination and cooperation represent the capacity to process information and influence the effect of buffering on the design. Finally, organizational size considers the organizational level to examine resource power and maturity of processes, essential factors of quality management and project management (Handley and Gray, 2015).

3. Research methodology

3.1 Sample and data collection

For testing the hypotheses, a survey method was employed, which collected quantitative data from 234 auto-component design projects in 124 automobile design organizations in China. The design agencies are, in nature, first-tier suppliers to fulfill the component design task of the buyer firm by integrating second-tier suppliers (Wilhelm *et al.*, 2016). Therefore, after obtaining an initial set of 497 automotive parts suppliers with the cooperation of a trade organization for vehicle manufacturers, the research team identified 124 available design agencies that had a recent experience with buyers on design projects and were willing to complete our survey. To minimize the threat of common method variance (CMV), the survey employed a multi-informant approach, and project managers and product managers in the design agencies are respectively selected to evaluate the independent and dependent variables. There is often one product manager position, but many project manager positions within a design agency. Product managers have strategic responsibility for driving the development of products, whereas project managers are responsible for overseeing the execution of those development plans (Becker and Zirpoli, 2003). Therefore, the product manager is more responsible for the outcome (design quality defined in this study) of the designed component, and the project manager is more responsible for the development processes of the component. During the development processes of components, project

managers continuously evaluate the context contingencies to represent the component project in negotiations with the buyer to promise the expected component to be delivered in limited time, and cost. Thus, project managers are suitable to serve as informants for the construct related to the project processes (boundary activities and context contingencies), and product managers in the design agencies are more suitable as respondents to evaluate the design quality (dependent variable).

Before the formal survey, for these 124 design agencies that agreed to join our investigation, their product manager (one product manager for one design agency) was first contacted and requested to identify up to three projects of component design for a single buyer. The project manager contact for each identified project was requested. After that, we began the formal survey to deliver the independent variable (including questions for the two moderators) instrument to the project managers, and dependent variable instrument to the product managers. In consideration of the long-term cooperative relationship between the project managers and product managers, we used a confidential survey in which only the research team could match survey instruments. In the 124 participating design agencies, 245 projects were identified by this process. After deleting 11 unreturned or incomplete responses, 234 usable observations are used for further data analysis. In the data, 55 design agencies reported 1 project within the product, 28 reported 2 projects, and 41 reported 3 projects. [Table II](#) provides the demographics of the research sample.

3.2 Construct development

Measurement items were based on the existing literature and adapted to our context. The instruments were extensively validated through initial feedback from design project managers, academic experts, and analytical post-collection validation. The initial items measuring buffering to protect from interference as OIB activity and buffering to prevent leaking as IOB activity are adapted from the scale of Ancona and Caldwell (1990). Initial items for environmental uncertainty are derived and adapted from the scale of the [Waldman et al. \(2001\)](#) and [Zhang et al. \(2012\)](#). The work of [Cooper et al. \(1998\)](#) provided initial items to measure internal project complexity. Finally, the auto-component design quality derives from items of [Nidumolu \(1996\)](#) and [Narayanan et al. \(2011\)](#). All measurement items designed to tap into the model's five constructs were pilot-tested on a group of project managers covering a wide variety of auto-component NPD projects in the automobile manufacturing industry. The final measurement items, as modified for our constructs, appear in [Table III](#).

3.3 Measurement assessment

Confirmatory factor analysis using structural equation modeling (SEM) techniques with LISREL 8.7 estimated the measurement model. The items to measure the corresponding

Table II.
Sample demographics

Organization size (employee numbers)		Capital on stock (10,000 RMB)		Project managers' age	
	%		%		%
<50	22.6	≤200	20.5	≤30	5.6
50–500	50.0	200–1000	39.4	31–40	12.0
501–1000	21.8	1000–2000	17.9	41–50	14.1
>1000	5.6	>200	22.2	>50	68.3
Organization age (years)				Project managers' position	
	%				%
≤5	13.2			Group leader	8.5
6–10	35.9			Section manager	8.5
11–15	29.5			Middle manager	44.9
16–20	4.3			Senior manager	38.1
>20	17.1				

Table III.
Measurements of the
constructs and
convergent validity

Constructs	Measurement items (and item-total correlation)
Outside-in buffering (Cronbach's $\alpha = 0.859$; confidence interval (CI) = 0.825,0.888; composite reliability = 0.911)	The project manager conducts the following activities (1—strongly disagree; 5—strongly agree): (1) Deflect outside pressures on the team (0.85) (2) Shield the team from outside interference (0.88) (3) Prevent overloading the team with external information and requests (0.74)
Inside-out buffering (Cronbach's $\alpha = 0.870$; CI = 0.830,0.896; composite reliability = 0.905)	The project manager conducts the following activities (1—strongly disagree; 5—strongly agree): (1) Withhold news about the team until an appropriate time (0.85) (2) Avoid releasing product information to others in the company (0.80) (3) Release only information that presents the desired profile (0.85)
External environmental uncertainty (Cronbach's $\alpha = 0.792$; CI = 0.742,0.888; composite reliability = 0.877)	Please evaluate the following as related to the environmental uncertainty of the component design project (1—strongly disagree; 5—strongly agree): (1) The external environment (economic/social/political/technological aspects) within which our project is very risky (0.77) (2) The external environment within which our project is very dynamic, changing rapidly in technical, economic, and cultural dimensions (0.66) (3) The external environment of our project is stressful, exacting, and even hostile (0.84)
Project internal complexity (Cronbach's $\alpha = 0.873$; CI = 0.842,0.899; composite reliability = 0.922)	Please evaluate the following as related to the project complexity of the component design project (1—strongly disagree; 5—strongly agree): (1) The different tasks within our project require different expertise (0.83) (2) The different tasks within our project greatly differ in the assigned schedule (0.83) (3) The different tasks within our project greatly differ in scope requiring the crossing of organizations (0.85)
Component design quality (Cronbach's $\alpha = 0.766$; CI = 0.713,0.811; composite reliability = 0.847)	Please evaluate the following as related to the final quality of the designed component (1—strongly disagree; 5—strongly agree): (1) In terms of product function, the component design meets requirements (0.64) (2) In terms of product performance, the component design meets requirements (0.64) (3) In timeliness, flexibility, and operability, the component design meets expectations (0.71) (4) In terms of overall product quality, we are satisfied with the component design (0.69)

constructs are suitable because all item loadings were above 0.4 and statistically significant (Bowen and Guo, 2011). The ratio of item loadings to their respective standard errors all exceeded the absolute value of 3.0 ($p < 0.01$). A power level of 0.8, with an alpha value of 0.05, was specified to calculate the minimum required sample size that satisfies the anticipated effect size (MacCallum *et al.*, 1996). Under these conditions, 138 is identified as the minimum

required sample size for the model structure. Therefore, the 234 observations in this study satisfied the minimum required sample size.

Furthermore, the low normed chi-square and good overall fit suggested by absolute and incremental measures of fit were exhibited in CFA results. Following [Shah and Goldstein \(2006\)](#), the basic measure of absolute fit is the chi-square statistic. The p -value of the chi-square test (chi-square = 153.27; df = 94; p = 0.000) is statistically significant. However, the chi-square test is inherently biased when the sample size is large as the chi-square statistic is very sensitive to sample size and overestimates the lack of model fit ([Bollen, 1989](#)). Thus, we adjust the chi-square statistic by its degrees of freedom, to make chi-square statistic less sensitive to sample size. Previous researchers recommend the ratio of chi-square to degrees of freedom (the normed chi-square) be less than 3 ([Bollen, 1989](#)) to indicate a reasonable fit. The normed chi-square in the CFA is 1.63, which indicates a good fit between the hypothesized model and the observed data. Further, [Browne et al. \(2002\)](#) indicate that GFI increases as does goodness-of-fit (bounded above at 1), while RMSEA and SRMR decrease as goodness-of-fit increases (bounded below at 0). The results of the CFA on absolute measures of fit are GFI = 0.92, RMR = 0.028, SRMR = 0.055, and RMSEA = 0.052. Incremental fit indices assess absolute or parsimonious fit relative to a null model. Incremental indices supplement absolute indices to add confidence to the overall fit of the model ([Shah and Goldstein, 2006](#)). For our model, the normed fit index (NFI = 0.92), incremental fit index (IFI = 0.96), and comparative fit index (CFI = 0.96) provide statistical evidence showing the underlying fit of the data from multiple perspectives. Consequently, the various indices of goodness-of-fit indicate the model to be an acceptable representation of the hypothesized constructs.

Construct reliability was evaluated by examining Cronbach's alpha, composite reliability, and corrected item-total correlations of each construct. Cronbach's alpha is a point estimate of reliability. However, the value of Cronbach's alpha may over- or underestimate the reliability of a latent variable because its measurement error terms may correlate, and measurement items are not essentially tau-equivalent. Further, the sample value of Cronbach's alpha may contain sampling errors of unknown direction and unknown magnitude ([Bonett and Wright, 2015](#)). Thus, a confidence interval for the population value of Cronbach's alpha is reported along with the sample value, because a 95 per cent confidence interval (CI) may have a lower limit that indicates poor reliability ([Bonett and Wright, 2015](#)). As it is only an estimate of reliability, the error of Cronbach's alpha should be considered through two-sided interval estimation ([Iacobucci and Duhachek, 2003](#)). To obtain the confidence intervals, we computed the Cronbach's alpha standard error for the point estimates using the procedure defined by [Iacobucci and Duhachek \(2003\)](#). Computation of such intervals allows an estimate of bias due to measurement error. Next, composite reliability was computed for each construct, as previous studies acknowledged this approach provides better estimates of reliability and overcomes the limiting assumptions of Cronbach's alpha ([Peterson and Kim, 2013](#)). [Table III](#) contains a summary of the results.

In [Table III](#), as a final consideration of convergent validity, item-total correlation coefficients are considered because they determine how much one item contributes to the internal consistency of the construct. To ensure measurement items are ideally correlated and have internal consistency among the items, we calculated corrected item-total correlations to serve as proxies for the correlations between the items and the latent variable. All measurement items have an item-total correlation of at least 0.64, and all were statistically significant at the 5 percent level. Considering the Cronbach's alpha confidence intervals, composite reliability, and corrected item-total correlations that indicate the construct captures the majority of variance, we are confident of construct reliability and convergent validity.

Discriminant validity focuses on whether the measures of constructs are distinct from each other. We used chi-square difference tests comparing two models for each of the ten

pairs of the five constructs to assess the discriminant validity. For the first CFA model, the correlation among the constructs was not limited and allowed to vary. In contrast, the correlation in the second model was fixed to a value 1.0. A Bonferroni corrected p -value of 0.1139 was used, given that several repeated tests are conducted (Kroes and Ghosh, 2010). Finally, we compared ten pairs of chi-square difference tests and found them all significant (p -value < 0.01), which promise the discriminant validity of constructs examined in this study. Finally, descriptive statistics and correlations for the variables are shown in Table IV.

3.4 Hierarchical regression analysis

In this study, hierarchical regression analysis was used to test the hypotheses. Given all five constructs in our research model are latent variables and each construct was measured by three or more observable items, we first computed the factor score for each construct using original data before we conduct regression analysis. In this study, during the LISREL 8.7 confirmatory factor analysis (CFA) to evaluate the measurement model, the factor scores of all constructs are computed by the software. These factor scores are exported to conduct the regression analysis. In Model 1, we entered all of the control variables into the regression model. Then, OIB and IOB as independent variables were added into Model 2 to examine their main effects on design quality. In Model 3, we entered the first moderator (external environmental uncertainty) and its interaction with OIB to examine the moderating effect of environmental uncertainty on the relationship between OIB and design quality. In Model 4, the second moderator (internal project complexity) and its interaction with IOB were entered to examine project complexity's moderating effect on the relationship between IOB and design quality. All variance inflation factor (VIF) scores for each construct in each model were below 2, which suggests that multicollinearity was not a concern in the analysis. Further, we examined the presence of outliers in order to reduce the possibility that a small number of extreme values would overly influence results (Cohen *et al.*, 2003). We calculated Cook's distance values for all cases and found they were all below 0.6, suggesting that extreme values do not influence our analysis.

4. Results

Table V shows the regression results. Results from Model 2 show that both the coefficients of OIB and IOB are significant for component quality improvement (OIB: $\beta = 0.148$, $p < 0.05$; IOB: $\beta = 0.211$, $p < 0.05$). Results of Model 3 show moderation of environmental uncertainty on the relationship between OIB and component quality is positive and significant ($\beta = 0.125$, $p < 0.05$). Finally, Model 4 shows that the interaction of internal project complexity and IOB has a significant effect on design quality ($\beta = 0.136$, $p < 0.05$). All hypotheses are statistically supported.

The PROCESS plug-in for SPSS details the conditional effect of internal project complexity and external environmental uncertainty. Results show that only when there is high environmental uncertainty, OIB contributes to the final design quality improvement

	Mean (Std.)	1	2	3	4	5
Protect from outside-in interference (OIB)	4.23(0.83)	1				
Prevent inside-out leakage (IOB)	4.15(0.56)	0.01	1			
Environmental uncertainty	3.37(0.59)	0.19*	0.08	1		
Project complexity	3.78(0.77)	0.12	-0.02	0.09	1	
Component quality	3.70(0.49)	0.16*	0.22*	0.40*	0.31*	1

Note(s): * $p < 0.01$, sample size = 234

Table IV.
Descriptive statistics

Table V.
Results of hierarchical
regression analysis

	Design quality				Std. Err adjusted for 124 design agency Model 5
	Model 1	Model 2	Model 3	Model 4	
<i>Control variables</i>					
Work age of project manager	−0.001(0.06)	−0.026(0.06)	−0.033(0.05)	−0.014(0.05)	−0.040(0.06)
Project team boundary spanning	0.085(0.08)	0.072(0.08)	0.048(0.08)	0.050(0.07)	0.035(0.05)
Project internal coordination	0.100(0.09)	0.064(0.06)	0.059(0.05)	0.069(0.05)	0.067(0.72)
Project internal cooperation	0.110(0.08)	0.141*(0.09)	0.113(0.09)	0.110(0.08)	−0.039(0.05)
Project size	0.074(0.10)	0.063(0.10)	0.055(0.09)	0.031(0.09)	−0.053(0.08)
Organization size (number of employees)	0.132*(0.08)	0.075(0.08)	0.083(0.08)	0.101(0.07)	0.066(0.07)
<i>Independent variable</i>					
OIB		0.145*(0.07)	0.119(0.07)	0.079(0.06)	0.086(0.06)
IOB		0.211*(0.07)	0.171*(0.06)	0.140*(0.06)	0.151*(0.06)
Environmental uncertainty			0.376*(0.06)	0.367*(0.06)	0.376*(0.06)
OIB × Environmental uncertainty			0.148*(0.06)	0.133*(0.05)	0.121*(0.05)
Project complexity				0.279*(0.06)	0.282*(0.06)
IOB × Project complexity				0.135*(0.06)	0.122*(0.06)
Adjust R^2	0.025	0.080	0.218	0.301	Adjust $R^2 = 0.282$
ΔR^2	0.025	0.055*	0.138*	0.083*	
Note(s): * $p < 0.05$, Standard errors in parentheses					

($\beta = 0.19$, $p < 0.05$, $SE = 0.09$, $LLCI = 0.004$, and $ULCI = 0.369$). When there is low environmental uncertainty ($\beta = -0.026$, $p > 0.05$, $SE = 0.07$, $LLCI = -0.161$, and $ULCI = 0.109$), the effect of OIB is not significant anymore. For IOB, we also find that internal project complexity is an important contingency for the effect of IOB on final design quality. The effect is not significant when project internal complexity is low ($\beta = 0.029$, $SE = 0.09$, $p > 0.05$, $LLCI = -0.147$, and $ULCI = 0.207$), but also only when project internal complexity is high ($\beta = 0.274$, $SE = 0.067$, $p < 0.05$, $LLCI = 0.141$, and $ULCI = 0.407$).

We conducted further steps to assess the robustness of our results. Specifically, we first used organizational capital as an alternative measurement of firm size and reran the analysis. Second, given the design quality is evaluated subjectively by the product manager, which may potentially expose our research results into the alternative explanation of social desirability response bias (SDB), we use the returns on assets as an alternative variable of design quality and rerun the regression model. No research findings changed considering either alternative. Finally, considering that the 234 surveys were received from 124 design agencies, the analysis should account for the project data clustered by design agency, as there may be unobservable errors within each design agency affecting the results. Therefore, we reran our research model clustered by the 124 design agencies and show the results as Model 5 in Table V, which indicates the research findings in this study are not threatened by the unobservable errors within the design agency.

5. Discussion

This study adopts a bidirectional view of team buffering work to investigate the design quality consequences of two boundary buffering activity sets (OIB for protecting against outside interference, and IOB to prevent confidential information leakage) in the context of a design team within a design agency's automotive component design project. Moreover, the effectiveness of buffering is considered for different conditions of external environmental uncertainty and project internal complexity. Based on survey data from 234 auto component design projects in 124 automobile design agencies in China, this study found that OIB activity contributes to the final design quality improvement when environmental uncertainty is high. Similarly, IOB activity increases design quality when internal project complexity is high. Drawing on these findings provides important theoretical contributions on team boundary management, research implications for project boundary management and quality management literature, and practical implications for design agencies in the automotive industry.

5.1 Theoretical contributions and implications

First and foremost, our work conducts a fine-grained investigation on boundary buffering management and clarifies boundary buffering as bidirectional work, namely, OIB and IOB, which contribute to the boundary management theory with a bidirectional view. Previously, buffering work was seen as a one-dimensional construct with no connotation of differences due to the direction (Faraj and Yan, 2009; Yan and Louis, 1999; Lynn, 2005). However, bidirectional consideration of flow accounts for boundary target locations that indicate buffering should also be bidirectional. In this study, the OIB buffering focuses on external actors (e.g. buyers) as targets to control the inbound information flow, but IOB buffering on internal actors (project members) as targets to filter the outbound information flow. Their boundary objects also differ. Recognition of these distinctions between inside-out and outside-in directionality should play a greater role in studies of boundary activity. Future researchers could further apply the bidirectional view to different boundary activities. Also, the six properties of boundary management theory summarized in this study provide a frame

for boundary management researchers to apply theory, which helps elaborate boundary activities in different contexts.

In addition, this study contributes to boundary management literature in three functional ways. First, prior project scholars recognize the importance of project boundary objects for project success but pay less attention to other boundary properties, especially the boundary targets location (e.g. [Koskinen and Mäkinen, 2009](#); [Iorio and Taylor, 2014](#)). This study shows the importance of considering more boundary properties to explain the effectiveness of the boundary activities, for example, the boundary target location and project complexity source identify different contingencies for appropriate buffering. Second, project scholars tout the relevance of both boundary spanning and buffering in project management, with strengthening critical to forming essential team drive and spanning detracting from required cohesiveness ([Lundin and Söderholm, 1995](#)). Others find the opposite, where new product development projects following an agile approach emphasize the importance of boundary spanning to project success ([Chakkol et al., 2018](#); [Prajogo and Sohal, 2013](#)) and a view maintaining that boundary strengthening will be detrimental to performance because the team works in an “isolation” condition ([Dey and Ganesh, 2017](#)). However, this study empirically demonstrates that both OIB and IOB have positive influences on the outcomes of design quality, addressing concerns about the effectiveness of the boundary-strengthening activities ([Dey and Ganesh, 2017](#)). Third, uncertainty is considered an important condition for the effectiveness of boundary spanning ([Faraj and Yan, 2009](#); [Gibson and Dibble, 2013](#)). However, this study shows that external environment uncertainty will also improve the effectiveness of OIB; otherwise, the design agency underinvests in quality ([Handley and Gray, 2015](#)). The findings open the possibility that targeting boundary buffering appropriately along both direction and target can allow for beneficial information intake while preventing interference or that promotional activity might occur while still preventing leakage. The approach taken allows greater consideration by researchers to examine the complementarity of strengthening and spanning activities ([Drach-Zahavy and Somech, 2010](#); [Yan and Louis, 1999](#)).

Finally, this study provides important implications to the inter-organizational quality management (QM) literature, especially in the context of the automotive component design agency as a third party inserted into the traditional buyer–supplier dyad. QM scholars are increasingly calling to depart from the perspective implied by most empirical QM literature—for researchers to examine contingencies which may positively moderate the effectiveness of QM practices in different contexts ([Sousa and Voss, 2002](#)), such as firm size, program duration, unionization, and industry ([Jayaram et al., 2010](#)), environmental uncertainty, and organizational structure ([Zhang et al., 2012](#)). These studies added nuance to the QM literature, yet they notably focus on managing quality at a single firm rather than across organizational boundaries ([Handley and Gray, 2015](#)). As more and more activities are outsourced, another key gap in the QM literature is the lack of direction about managing quality across a supply chain ([Handley and Gray, 2015](#)). Accordingly, this study offers a unique contribution to the QM literature by considering the inter-organizational OIB of the design project as inter-organizational QM practices, and identified external environment uncertainty as contingencies to effectiveness. Important results indicate that the buffering work should receive more attention in the inter-organizational quality management field, particularly regarding the flexibility expected by the buyer. Future studies from the perspective of the buyer should take conflicts between the two actors into account, primarily due to the uncertainties present that require design changes ([Stevenson and Spring, 2009](#)).

5.2 Practical implications

Design agency teams need to pay more attention to boundary buffering, which is sometimes ignored or insufficiently conducted but which plays a vital role in contributing to better automotive component design quality. An open door to an inflow of information may bring in

more knowledge and opportunities but ignore the impact of interference resulting in overloading, altered plans, or talent raids. Therefore, practitioners must balance boundary spanning and boundary strengthening. Such advice also applies to outflow considerations. Boundary-spanning activities expose a team to the potential loss of sensitive information. With regards to the consequences of beneficial spanning, our findings remind a project manager to simultaneously strengthen both outside-in and inside-out boundaries when uncertainty and complexity are evident. Practices to implement buffering in both directions still require establishment, as well as metrics to determine the efficacy of practices. Our research findings also provide implications for the buyers who want to maximize design quality: it is better to provide extra incentives for quality efforts of the design agency.

5.3 Limitations

Two limitations of this study should be noted. First, this study does not consider the dynamic nature of a project, which has been found by project scholars to explain different boundary activities required and changes to their relative importance during the project life cycle (Dey and Ganesh, 2017). It's expected that boundary buffering would be low at the beginning stage of component design so that the team is open to outside expertise and knowledge. However, at the later stages of the project, boundary buffering should be greater to avoid unnecessary and high-cost changes initiated by outsiders. Therefore, beyond the static perspective of the project management in this study, it will be interesting to conduct a longitudinal design with multiple measurement intervals as important extensions and to explore whether the intensity and types of boundary buffering activities will change over the stages in the life cycle of the project. Second, for the operationalization of project complexity, we did not capture all sources of complexity, such as "technology complexity" and "socio-political complexity" (Liu et al., 2019; Geraldi et al., 2011). Future research should consider a complete spectrum of complexity.

6. Conclusion

In the context of the automotive components industry, interference from outsiders to a design team and information leaking from the NPD project go against the effectiveness of the design work, which challenges the design project team to defend their boundary. Distinguishing the inbound and outbound information flow, a bidirectional view on the buffering work identified two kinds of boundary buffering, namely, OIB and IOB. Expectations that both significantly contribute to the final design performance were confirmed by analysis of survey data. Further, two contingencies (external environment uncertainty and internal project complexity) for the two buffering activities are identified and tested. The findings remind practitioners to treat the buffering work seriously in a mix of boundary activities. OIB should buffer against damaging interference, especially when the external environment is uncertain. Similarly, when project complexity is high, IOB is required to prevent information leakage. Research should consider the implication that boundary directions are important along with boundary targets, objects, activity, and function.

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